

(a) Benefit mechanism

Larger benefit when hit rate p is high and the speculator is faster ($\alpha \gg \beta$).

Sequential

Actor step t

Actor step $t + 1$

prefetched result
ready in cache

$$\text{reduction} \approx \frac{p}{1+p} \times \frac{\alpha}{\alpha+\beta}$$

Speculative

Speculator
predicts $t + 1$

Actor t

Verify
and commit

hidden wait /
time saved

time

■ Actor / tool step ■ Speculator ■ Hidden wait

Illustrative schematic derived from the dialogue summary; not to scale.